Fall Detection With UWB Radars and CNN-LSTM Architecture

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***Abstract*—Fall detection is a major challenge for researchers. Indeed, a fall can cause injuries such as femoral neck fracture, brain hemorrhage, or skin burns, leading to significant pain. However, in some cases, trauma caused by an undetected fall can get worse with the time and conducts to painful end of life or even death. One solution is to detect falls efficiently to alert somebody (e.g., nurses) as quickly as possible. To respond to this need, we propose to detect falls in a real apartment of 40 square meters by exploiting three ultra-wideband radars and a deep neural network model. The deep neural network is composed of a convolutional neural network stacked with a long-short term memory network and a fully connected neural network to identify falls. In other words, the problem addressed in this paper is a binary classification attempting to differentiate fall and non-fall events. As it can be noticed in real cases, the falls can have different forms. Hence, the data to train and test the classification model have been generated with falls (four types) simulated by 10 participants in three locations in the apartment. Finally, the train and test stages have been achieved according to three strategies, including the leave-one-subject-out method. This latter method allows for obtaining the performances of the proposed system in a generalization context. The results are very promising since we reach almost 90% of accuracy.**

***Index Terms*—Fall, Detection, Classification, Ultra-wideband radar, CNN-LSTM, Leave-one-subject-out.**

I. Introduction

RECENT demographic studies have statistically confirmed the overall aging of the world's population. For example, the median age increased from 21.5 years in 1970 to over 30 years in 2019 [1]. Besides, this aging is more significant in western societies where the elderly populations can reach meaningful proportion. Indeed, the medicine and quality of life have improved significantly, which affect directly the life expectancy (increased by a rate of 5 years from 2000 to now) [2]. For example, in Canada, the number of seniors increased from 3.88 million in 2001 to 16 million in 2016. Also, according to forecasts by the federal agency, one in four Canadians will be more than 65 by 2030, which would represent about 25% of the Canadian population [3]. Therefore, the issues associated with the care and well-being of older people are becoming major concerns for families, institutions, and companies. Currently, it is common for families to institutionalize their seniors in care centers. The main advantage of this solution is the continuous presence (24/7) of caregivers, nurses, and physicians. They can intervene in cases of discomfort, injuries, or meet the needs of the elderly. Nevertheless, there are still periods (during the day and night) where seniors are left alone. These periods are defined as critical because events such as myocardial infarction, respiratory distress, or fall can occur without anyone being informed. In cases like these, it is vital to intervene as quickly as possible because injuries provoked by these events can get worse and become permanent, leading to a painful end of life (e.g., physical and cognitive disability) or even death.

The work presented in this paper focuses on fall detection. Indeed, according to the World Health Organization (WHO), 30% of the persons over 65 years suffer accidentally from one or more falls per year, and for persons over 80 years, this rate increases to reach 50% [4]. This fact is more alarming when one considers that falls often happen in indoor environments and are related to normal activities of daily living (ADLs). The consequences related to a fall are major injuries (e.g., femoral neck fracture, brain injuries, skin burns), often leading to physical and cognitive disability or death in cases of undetected falls [5]. However, it is possible to reduce these risks (death rate and severity of injuries) by detecting falls as quickly as possible. In that sense, researchers proposed a significant number of solutions, which can be divided into two categories. The first one relies on the use of wearable devices [6], [7], such as smartwatches or smartphones. Despite the popularity of this approach, the number of false positives (identifying a fall when it is not) is consequent [8]. Also, it is frequent that seniors forget to wear the device, conducting the system to be ineffective [9]. Moreover, wearable devices have a limitation concerning battery management since there are inoperative if the battery is discharged. The second category is represented by non-wearable solutions, such as smart floors or cameras [10], [11]. Although non-wearable solutions represent an interesting alternative due to the absence of battery management and a non-dependency of seniors (wearing the device), the related work does not provide enough tests to evaluate performances correctly.

Despite the significant number of solutions to recognize falls in the literature, there are two main challenges. The first challenge is to develop and adapt various solutions (technologies) to

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senior's preferences. The main reason is that seniors can feel discomfort (well-being reduced) with a particular technology (e.g., cameras are considered intrusive), and hence, it is necessary to propose alternative solutions.

Thus, we propose a new system based on multiple ultra-wideband (UWB) radars to detect falls. The motivation to exploit this type of radar is the richness of information without the possibility of identifying the person or its activities as easily as images provided by cameras. Then, the last challenge is taking advantage of UWB radars and manipulating data to differentiate falls from ADLs. In addition, we programmed one architecture of deep neural network (DNN), which is a convolutional neural network (CNN) combined with a long short-term memory network (LSTM). Compared to the literature exploiting this type of sensor, the work presented in this paper gives new research results. Indeed, we conducted experiments with 10 participants, at three different locations in a realistic home setting of approximately 40 square meters. Only a few existing works collected data in such an environment (e.g., size of the apartment, fully furnished) [12], [13]. The data related to simulated falls and ADLs (e.g., cleaning a room, drinking a glass of water) have been acquired thanks to three UWB radars at different times in the day and different days (over two months). It should be mentioned that simulated ADLs are more complex than those in the literature. Also, it is the first time that three UWB radars have been exploited at the same time to detect falls. Indeed, the maximum number of radars that we found in the literature is two [14]. In addition, four types of falls, namely forward fall, side (left or right) fall, backward fall, and fall after trying to sit down, are simulated according to a video capture study [15]. Although there are four types of falls, the addressed problem is a binary classification attempting to differentiate fall and non-fall events. The falls are therefore based on documented cases in the scientific literature and the diversification make this study more realistic than most of those found in the literature. This resulting dataset is our first contribution. The second contribution represents the testing phase of the trained CNN-LSTM models. Specifically, we tested deep learning models in order to obtain performances which would generalize to different contexts. To do so, the dataset was split with three strategies. The first one was split on the instances. The second strategy separates files of the dataset. The last one is based on the leave-one-subject-out experiment, which is rarely used in existing studies. The idea is to divide the dataset into $N-1$ persons to train the model and test it with the Nth person. Applying these three strategies in the same work is also a novelty compared to related work. We obtained very promising results since the average accuracies on the third strategy are 88.1%, 91.5%, and 82.6% for positions 1, 2, and 3, respectively. In the best case, where we attempted to detect falls with the data provided by the closest UWB radar (only one radar), we reached accuracies of 98.51% (position 1), 96.75% (position 2), and 95.48% (position 3).

The remainder of the paper is organized as follows: Section II presents a complete state of the art dealing with fall detection systems. Section III describes the setup. More specifically, details of the apartment and UWB radars are given. Section IV explains the data preprocessing and CNN-LSTM architecture. In Section V,the experiments that have been conducted are detailed. Section VI introduces performances and discussion. Finally, the last section concludes the proposed work and gives future works.

II. Related Work

The literature on fall detection can be divided into two main trends, which are the techniques relying on wearable technologies and those relying on non-wearable solutions.

A. Wearable Solutions

For the first one (wearable devices), most of the researches exploit a triaxial accelerometer [6], [7] to detect falls. This sensor is integrated into a device (e.g., smartwatch, smartphones, box attached to a belt) that users can wear in some locations of the body. The most common locations are the waist [6], a wrist [16], or the chest [17]. In some cases, researchers also combine gyroscope [18] or air pressure sensor [19] into the device. The role of these technologies is to capture human movements by providing a series of values (raw signals). Then, signals are used to detect falls with different approaches divided into three detection strategies: (i) Threshold + If-Then rules; (ii) Feature Extraction + Machine Learning Algorithms; (iii) Raw Data + Machine and Deep Learning Algorithms. Strategy (i) [20], [21] is defined as a handcrafted method since it relies, in general, on the detection of the maximum or/and minimum peak(s) in signals. Then, the identified peak(s) is compared with a threshold value computed from the available dataset. Finally, rules (if-then) can be used to classify events. However, this strategy presents a significant number of false positives (disturbing the nurses in their work) and false negatives when the model is tested with new instances. For example, in [20], the rate of false negatives (falls classified as non-fall event) was 10.4%. The main reasons of this unsatisfying performance are related to the low number of features and the use of handcraft rules based on threshold values. Strategy (ii) [22]–[25] relies on features extracted (e.g., maximum and minimum peak, acceleration, angular velocity, velocity) from a sliding window and classical machine learning algorithms (e.g., k-Nearest Neighbors, Support Vector Machine, Nave Bayes). The performances are very good. For example, in [23], they obtained an area under the receiver operating characteristic curve (AUC) of 0.96. Concerning the strategy (iii) [7], [16], [17], it is based on the use of raw data provided by sensors. Indeed, the raw data represent the inputs of classical and deep neural network algorithms. However, the main disadvantages of this first trend are battery management, cases where users forget to wear the device, and some uncomfortable body locations (e.g., chest, back, waist).

B. Non-Wearable Solutions

For the second trend (non-wearable devices), the literature can be divided into two categories according to the used technologies. We divide them into those depending on vision sensors and those using non-vision sensors.

The first category is based on the use of cameras and computer vision techniques. Cameras are very popular with researchers

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since the information contained in the image is rich. However, unlike researchers, humans are reluctant to accept cameras due to privacy concerns. Thus, other types of cameras have been proposed to detect falls, such as thermal and depth cameras [10], [11], [25], [26]. The main advantage of these latter technologies is that patients cannot be recognized (or it is a very complex task to recognize patients). Indeed, these cameras do not provide details about the face and clothes, making identification almost impossible. However, the movements (actions) performed by the patients can be easily distinguished, which can still affect their acceptance of the technology. Concerning exploited methods to detect falls with cameras, the literature proposes innovative approaches.

For instance, Khan *et al.* [26] present an architecture of a deep neural network composed of one autoencoder followed by one 3D convolutional neural network. The model has been trained with only image sequences of ADLs (and by excluding fall events). The output of the model is a probability related to the chance of an image sequence to be an ADL. If the probability is close to 0, then the image sequence is probably a fall. They tested their approach on three open datasets and obtained the Area Under the Curve (AUC) values of 0.91 (twice) and 0.95. The approach is interesting but needs all existing ADLs in a real-case.

In another example of a vision-based approach, Lu *et al.* [27] proposed to detect falls from video kinematic data. The videos are provided by open datasets such as the Multiple Cameras Fall Dataset (MCFD), the Fall Detection Dataset (FDD), or the UR Fall Dataset (URFD) and so on. However, the authors mainly focused on MCFD. This dataset covers 24 scenarios, where each scenario is composed of nine activities. Also, eight cameras separately captured the scene in each scenario. It results in 192 (24x9) videos for a total of 736 actions. The frame rate of each video is 30 fps. They developed a 3D CNN combined with an LSTM to classify the data (Fall and Non-Fall actions). The 3D CNN is trained to extract discriminant features. In addition, the 3D convolution could extract motion features from a temporal sequence while the 2D CNN could only encode spatial information. The LSTM has the property to locate the region of interest in each frame. Hence, the proposed combination of the 3D CNN and the LSTM locates the region of interest in the video and encode the motion information. Also, it should be mentioned that 3D CNN and the LSTM have been separately trained. The video inputs are divided into 15 clips of 16 frames, and the amount of overlap between two consecutive clips is eight frames. They employed the five-fold cross-validation in order to validate their approach. The accuracy obtained reaches 100%, which is the best performance on the MCFD. The authors confirmed the efficiency of their approach by obtaining excellent results on the other datasets, namely the FDD and the URFD. Indeed, the accuracies are 99.36% and 99.27%, respectively. Finally, they transposed their deep learning algorithm on datasets specialized in video-based activity recognition. The performance varies from 50% to 98% approximately.

To conclude this part, in [28], the authors proposed a system detecting in real-time human falls in naturally occurring scenes. The system relies on a vision-based approach using cameras and a deep ConvNet. More specifically, their work is based on the concept of *dynamic image*, which combines the frames of a video into a single image in order to identify the sequences of action. Then, the deep ConvNet is trained with the dynamic images, and the final decision is given by a sliding window framework and a Difference Scoring Method (DSM). They evaluated their approach on three publicly available fall detection datasets (simulated falls) and on the YouTube Fall Dataset (YTFD). The performances have been expressed with the sensitivity and specificity, and the scores greatly depend on the dataset. For example, the sensitivity and specificity obtained on the Le2i fall detection dataset are 98.43% and 100%, respectively. For the YTFD, the values of the sensitivity and specificity are 63.7% and 68.1, respectively.

The second category relies on sensors providing one-dimensional (1D) discrete-time signals. Compared to cameras (providing 2D, 3D, and 4D data), the information contained in 1D signals is complex to interpret by an observer. Also, these technologies overcome the limitations of wearable devices and cameras, namely battery management problems, the absence of data (when patients forget to wear the device), and privacy concerns of cameras. According to the literature, the number of solutions to detect falls is extensive. Indeed, smart floors (where the floor is composed of a network of seismic sensors, accelerometers, or pressure sensors) [29], acoustic sensors (e.g., ultrasonic array sensors) [30], or infrared array sensors [31] are examples of technologies to detect falls.

During these last years, some researchers did exploit radars and machine learning algorithms to recognize falls and human activities [32]–[34]. Some of them used data provided by continuous wave (CW) radars [35], [36]; others take advantage of the information given by frequency-modulated continuous-wave (FMCW) radars [33], [37] and, in some cases, UWB radars were employed [14], [38]. Two different ways (uses) to perform this recognition are statistical learning [14], [33], and deep learning [37]–[39] approaches.

For example, Erol *et al.* [32] presented a method that exploits the slow-time, fast-time, and Doppler frequency (three radar signal variables) to generate a radar data cube (RDC). They then applied a multi-dimensional principal component analysis (MPCA) to reduce the number of information of the RDC. In other words, the MPCA acts as a feature extractor. The remaining data from the RDC were used to feed a k-NN algorithm. The accuracy is 97.88%, which outperformed the standard principal component analysis (PCA). The latter has only 86.01% for the accuracy. Unfortunately, the experiments are the weak part of the study. Indeed, only six participants took part in the experiments, and only four classes are considered, where each class has approximately 100 recorded motions of 10 s duration. Also, the approach has not been tested with the leave-one-subject-out strategy, which allows determining the real performances of the system with new persons (in a real case) and confirm generalization over new participants.

Another example, Shrestha *et al.* [33] proposed detecting falls and recognizing human indoor activities using radar systems. The main objective of this work is to show the potential effectiveness of various features and classifiers. Two datasets

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were created to reach this goal. The first one is composed of experiments performed in an indoor meeting room at the University of Glasgow. They used one FMCW radar located at the height of approximately 1.2 m to be in front (of the torso) of the participants. Six persons participated in the data collection. These persons simulated twice each of the seven activities. The second dataset has been created in the laboratory of the Telecommunication Systems Group at the Universit Politecnica delle Marche. The same radar was used. However, this dataset is composed of ten activities, including the simulation of a frontal fall after tripping. Seven participants took part in this experiment. Additional data were recorded for the same activities performed by two older people (58 and 62 years old). Different signal processing techniques were successfully applied (e.g., Short Time Fourier transform, IIR filter) on the radar data to generate micro-Doppler signatures. From those signatures, the authors extracted ten features (e.g., centroid and bandwidth of the signatures, the entropy of the signatures) to create vectors (instances) that can be used to classify data with classical machine learning algorithms. The best performances were obtained with the Support Vector Machine (SVM). The dataset has been randomly shared into a training (80% of the data) and a testing set (20% of the data), and they repeated 50 times this splitting strategy. They reached 94% and 76.44% (average accuracies) for the first and the second dataset, respectively. These results show the influence of the type of activities performed by the participants (activities in the second dataset are more complex than those in the first dataset) and the number of participants on the performances. It is noteworthy that the fall class in the second dataset has 82% accuracy, which is low compared to our paper.

Sadreazami *et al.* [40] used only one UWB radar and one CNN architecture to recognize fall events. They obtained an accuracy of 92.72%. However, the experiment presents some limitations. First, the dataset is poor in diversity, since there are only 5 participants. Also, all participants performed 121 falls and 85 non-fall activities. Secondly, the proposed approach has not been tested following the leave-one-subject-out method. In other words, the obtained performance is not representative of the generalization (for any subject) of the solution.

Another example, Wang *et al.* [41] also proposed to exploit one UWB radar to detect falls. The range-time image provided by the radar is processed with the short-time Fourier transform in order to get time-varying range-Doppler images. Then, a sequence of this latter image is used to extract features with a convolutional AutoEncoder. Finally, the extracted features fed a GRU architecture [42]. They reached an accuracy of 96.80%. Despite this interesting approach, the experiments also presents important limitations. Indeed, there are only four participants. Also, all activities have been performed in only one location. It shall be noted that all research, to the best of our knowledge, presented in the literature did not test the generalization of their approach with the leave-one-subject-out method, and recorded activities at only one location. It has been shown that the angle of the performed activity with the radar has an influence on the recognition results [34].

Li *et al.* [37] used a multi-layer bidirectional LSTM for multimodal sensor fusion to classify daily activities' patterns and falls. The data are acquired from three wearable inertial sensors (on the wrist, waist, and ankle) and one Frequency-Modulated Continuous-Wave (FMCW) radar. Each inertial sensor has one tri-axial accelerometer, gyroscope, and magnetometer. 16 participants simulated daily activities in a common room at the University of Glasgow. More specifically, the radar was placed in front of the activity zone of a surface size of 3 m $\times$ 2.2 m. The dataset is composed of six human activities (walking, sitting on a chair, standing up, bending to pick up an object, drinking a glass of water, and simulated a frontal fall). It should be noted that the authors take into consideration the transition between activities. To validate their approach, they employed the strategy of the leave-one-subject-out. The best results were obtained with the hybrid approach of soft-hard combination fusion. Indeed, they reached approximately 96% of average classification accuracy. However, the two drawbacks of this work are the experimental design (which is limited in a small activity zone), the daily activities (which are simple), and the difficulty of setting the soft-hard combination fusion (the number of virtual classifiers) correctly.

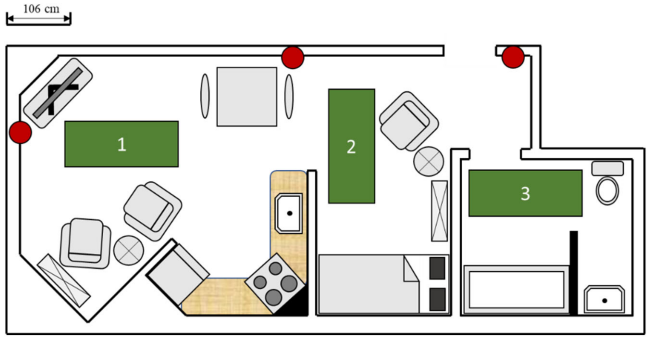


Fig. 1. Plan of the LIARA apartment. red circles are the location of the UWB radars. green rectangles are the location of simulated falls.

For all these reasons, the proposed study is interesting for the scientific community because it realistically depicts the real conditions for detecting falls and uses the leave-one-subject-out strategy to show the generalization over new participants.

III. Setup of the Study

The achieved study, namely, fall detection in an indoor environment, has been completed in the Laboratoire d'Intelligence Ambiante pour la Reconnaissance d'Activits (LIARA). The LIARA has an apartment of 40 square meters and possesses more than a hundred different sensors (e.g., radio frequency identification transponders and antennas, contact sensors, passive infrared sensors) and effectors (e.g., screens, speakers, lights). The objective of these sensors is to provide data and enhance the habitat with ambient intelligence. Fig. 1 shows the plan of LIARA's apartment.

In this study, the proposed fall detection only relies on the three UWB radars installed in the apartment.We decided to consider using UWB radars instead of CW radars due to better penetrating and fine range resolving abilities of UWB radars [43], [44]. Red circles in Fig. 1 give their locations,

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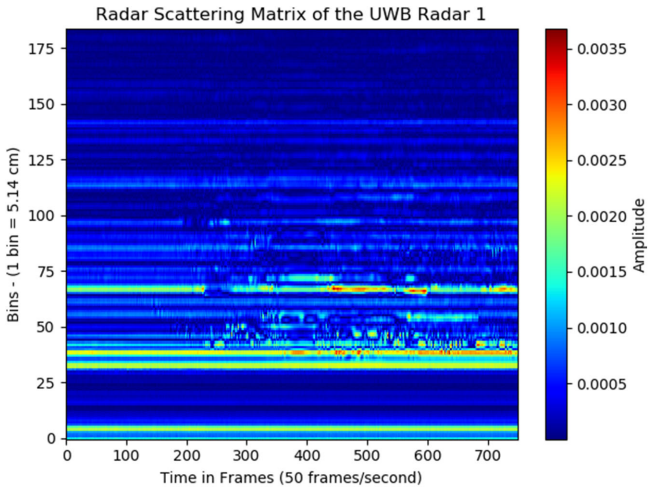


Fig. 2. The radar scattering matrix provided by one UWB radar during a non-fall activity.

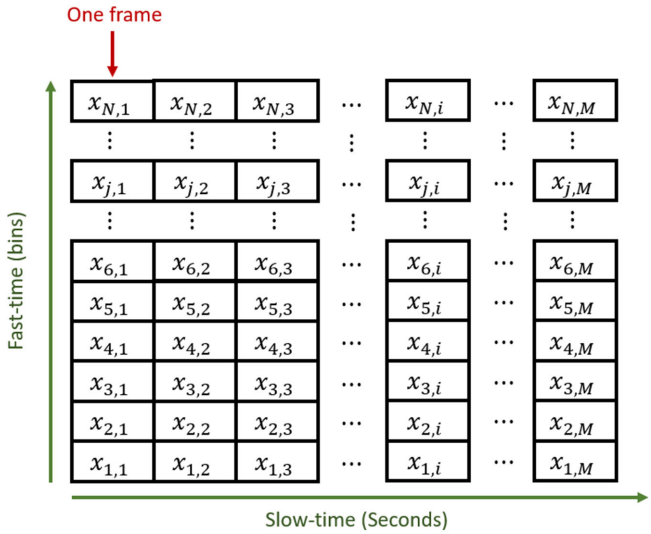


Fig. 3. The radar scattering matrix.

and each of them is mounted 36 cm above the ground on the wall. This position above the ground has been chosen because electric outlets, in Canada, are at this height. The idea is to have a plug-and-play solution. Also, we exploited the Xethru X4M200 from Novelda. This radar uses a UWB transceiver operating in the frequency range of 6.0–10.2 GHz (6.0–8.5 GHz for low frequency 252 bands, and 7.25–10.2 GHz for high frequency bands), which provides high spatial resolution, unobtrusiveness, and privacy-preserving monitoring by sending short pulses at high frequencies. These radars have been configured to acquire scans (frames) at a rate of 50 frames/second. Thus, recorded data during a period P results in a radar scattering matrix, where each row represents the spatial samples from different ranges (bins – fast time), and each column represents observations (frames) recorded at different time intervals (slow-time). Fig. 2 illustrates one radar scattering matrix. Also, the scanning system has been set to operate in a distance range of [0.4–9.8] meters implying that one bin corresponds to 5.14 cm.

IV. The Proposed Method

The proposed method can be decomposed into two steps. The first step is the data preprocessing to transform the radar scattering matrix into a time series. The second step is to exploit a CNN-LSTM architecture in order to generate classification models.

A. Data Preprocessing

Let $\mathbf{X} = x_{j,i} \in \mathbb{R}^{N \times M}$ be the radar scattering matrix in a conceptual form illustrated in Fig. 3, where $x_{j,i}$ is the amplitude value related to the bin j of the frame i. N and M denotes the total number of bins of one frame (fast-time) and the total number of frames (slow-time) respectively. In other words, $\mathbf{X}$ represents a successive concatenation (recording) of the received radar signals (frames) during a specific period (time window), say P seconds. Thus, M corresponds to the multiplication of the sampling rate (S frames/second) by the duration of P seconds. Then, the data preprocessing step has the objective to transform this radar scattering matrix $\mathbf{X}$ into a radar time serie $\mathbf{y} = y_i \in \mathbb{R}^{1 \times M}$. To do so, we normalize all the data in one column (amplitude values of one frame) according to the maximum absolute value in this same column. Then, we sum all the normalized values of the column in order to obtain the value y_i at time i of our new signal. Equation (1) describes the operation transforming the radar scattering matrix into a radar time-series.

$$y_i = \frac{1}{\max_{1 \leq j \leq N} (|x_{j,i}|)} \sum_{j=1}^{N} x_{j,i}, \forall i \in \{1, 2, \ldots, M\}. \quad (1)$$

Finally, the resulting time-series $\mathbf{y}$ is also normalized by subtracting the mean and dividing by the standard deviation in order to provide the time-series $\mathbf{z} = z_i \in \mathbb{R}^{1 \times M}$. Equation (2) expresses this last operation to obtain a time-series that is the input to deep learning models.

$$z_i = \frac{y_i - \text{mean}(\mathbf{y})}{\text{std}(\mathbf{y})}, \forall i \in \{1, 2, \ldots, M\}. \quad (2)$$

The resulting time-series represents one instance of the dataset used to train the deep learning models. Fig. 4 shows the normalized radar time-series of one activity.

It shall be mentioned that the time window defined in this work has a length of 15 seconds, with an 95% overlap between each window. The choice to set 15 seconds for the time window length was made according to the experiments (falls and the types of the simulated ADLs). Indeed, in the case of the falls, data have been recorded for 20 seconds. More specifically, the participants started to simulate the fall between 5 and 10 seconds from the beginning of the recording. Then, the participant simulated the fall and stayed on the mattress with the possibility to move until the end of the recording. We wanted to get the maximum of the fall sequence in the time window to get better performances. In addition, the performed ADLs (e.g., washing your hands, cleaning the room, taking the medications) are well recognized with

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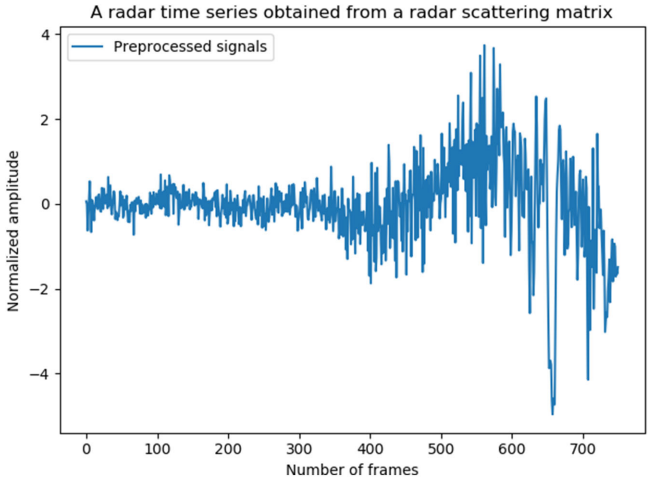


Fig. 4. Resulting time series from the data preprocessing applied to the radar scattering matrix of a non-fall event.

a larger time window. Hence, the length of 15 seconds for the time window was satisfying in order to differentiate falls from ADLs. For the overlap between two successive time windows, the choice was made according to the response performance of the system that we want. We wanted, at least, one classification per second. With an overlap ratio of 0.95, the system can provide a classification every 0.76 second, which is also satisfying.

Since we have three UWB radars, the same preprocessing is applied to the two other radars. Consequently, one real instance is composed of three time-series denoted by $[TS_1, TS_2, TS_3]^T$, where TS_1, TS_2, and TS_3 are the time-series related to the UWB radar 1, UWB radar 2, and UWB radar 3, respectively.

B. CNN-LSTM

The architecture of the CNN-LSTM network exploited in this work is constructed by stacking 1D CNN, two LSTM layers, and one fully connected layer. The choice to exploit the architecture of the CNN-LSTM was made according to the preliminary results obtained with an LSTM model. The LSTM model has the same architecture as the LSTM part of the presented CNN-LSTM. In the case where we only used the data of one radar related to its closest fall, we obtained 89%, 85% and 76% of accuracies for the position 1, 2, and 3, respectively. Also, the processing time of the training stage is much longer than the CNN-LSTM. Indeed, the LSTM model trained for 100 minutes approximately while the CNN-LSTM architecture trained for 2 minutes and 30 seconds. Hence, The CNN-LSTM model can automatically extract features (better than an LSTM model) and learn time-series patterns for classification. It is the main reason for this deep learning model choice. Fig. 5 illustrates the architecture of the CNN-LSTM. However, we must make these three types of neural networks compatible. Indeed, since one fall is an event evolving dynamically with time, some temporal relationships from the signal (time series) should be extracted. Thus, the input (pre-processed data) of the CNN-LSTM model is split into five subsequences, where each of them has 150 samples. Then, each subsequence feds one CNN

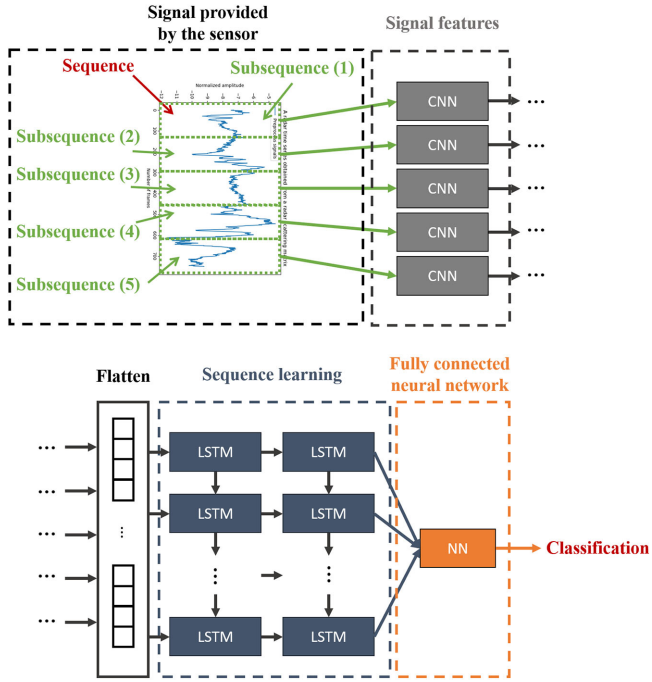


Fig. 5. The CNN-LSTM architecture.

to extract relevant information. It results in five series of features feeding the LSTM network. LSTM network is a special kind of recurrent neural network learning from short-term as well as long-term dependencies. In other words, LSTM networks can model the temporal changes in a sequence of data thanks to their recurrent architecture and memory units [45].

More specifically, the 1D CNN provides an abstract representation of the input sensor data in feature maps thanks to the convolutional filters. It shall be mentioned that we added a dropout (rate of 0.5) layer between each layer of the CNN-LSTM model. Dropout layers consist of setting to 0 a fraction of input units in order to prevent overfitting. The CNN architecture contains two one-dimensional convolutional layers: 64 (1×3) and 64 (1×3) filters respectively in the forward direction. The two parameters consist of the definition of the dimensionality of the output space and the length of the 1D convolution window, respectively. Each of this convolutional layers implements the rectified linear unit (ReLu) activation function. The other parameters are those set by default in KERAS.[1] After these convolutional layers, we added a one-dimensional max pooling layer with a pool size equal to 2. This layer allows to down-sample an output with also the objective to prevent overfitting. After the CNN part, the output is flattened into a one-dimensional vector feeding the LSTM network. As explained, the role of the LSTM network is to recognize and synthesize temporal dynamics. In this CNN-LSTM architecture, the LSTM network has two stacked layers. The choice of stacking layers of LSTM units has been made according to the work presented by Grave *et al.* [46]. Indeed, he demonstrated that the depth of the network is more important than the number of memory cells. Stacking LSTM

[1]https://keras.io/

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layers allows extracting better information than a single layer. Hence, the first layer has 100 LSTM units, and the second one has 50 LSTM units. The decrease of LSTM units between the two LSTM layers reduces the chance to explode the dimensionality, leading to an overfitted model. To conclude the architecture of the CNN-LSTM, we connected the output of the LSTM to a fully connected neural network having two layers. Layer 1 has 100 neurons set with the ReLu activation function. Layer 2 is defined by the SoftMax activation function in order to classify instances in *Fall* or *Non-Fall* class. It is noteworthy that the objective of this work is not to propose the perfect architecture. Therefore, we did not perform a grid search to determine the best hyperparameters of the presented model.

V. Experiments

All experiments have been conducted in the LIARA apartment under ethical clearance. Indeed, ethical clearance was obtained through the Universit du Qubec Chicoutimi human research ethics board (ID number: 2019-235). The study implied ten male participants, aged from 23 to 39. Also, all participants were healthy adults. It shall be mentioned that data acquisition has been achieved at different time in a days and different days (over two months).

The procedure of the data acquisition has been designed to demonstrate that falls can be detected from UWB radars. More specifically, we wanted to obtain performances that showed the proposed system operating under realistic conditions. To do so, each participant simulated four types of falls, namely forward fall, side (left or right) fall, backward fall, and fall after trying to sit down.

The simulated fall can be divided into two steps. In the first one, the participants simulate the fall between 5 and 10 seconds. In the second step, the participant stays on the ground and does not move much anymore. The overall period to perform one fall (the two previous steps) was 20 seconds and corresponded to the time of the data acquisition. Each type of fall was repeated five times at three positions in the apartment, as illustrated by green rectangles in Fig. 1. Thus, the total number of simulated falls is 600. All participants performed one time two minutes of ADLs for each batch of simulated falls (20 falls per position), resulting in a total of six minutes of simulated ADLs per participant. During these six minutes, participants were free (without constraints) to perform any ADLs anywhere in the apartment. Finally, we have 200 minutes of recording data for *Fall* and 60 minutes of *Non-Fall* data. The dataset is unbalanced if we considered the number of minutes. However, we created instances by applying a sliding time window. To do so, we defined 15 seconds for the length of the time window with an overlap of 95%. Consequently, the *Fall* class and *Non-Fall* class have 140 instances each (7 instances x 20 falls) per position per participant.

VI. Results and Discussion

In general, a classification model is generated (trained) with a training dataset, and evaluated by using a testing dataset. Then, we compute some well-known metrics representing the performances of the classification model. In this study, we exploited the accuracy (Acc.), Kappa measure (K.), precision (P.), the recall (R.), and the F1-score (F1-Sc.).

Moreover, to propose an in-depth study, we decided to explore three strategies to evaluate the performances. The first strategy consists of splitting the dataset (all created instances) randomly into two parts, where 70% of instances represent the training dataset, and the balance (30% of instances) defines the testing dataset. As described previously, one instance is given by three time-series (one per UWB radar) on a period of 15 seconds (time window). Also, we defined an overlap of 95% between two time-windows. In other words, one data file consisting of a recorded event over 20 seconds has multiple instances (7 instances can be created with an event of 20 seconds). This splitting method is the traditional way of evaluating classification models in machine learning.

The second strategy focuses on file splitting. More precisely, 70% of files constitute the training dataset, and the testing dataset is composed of the rest (30% of files). The main difference with the first strategy is that the split has been performed before the creation of instances. Indeed, in the first strategy, we created instances from data files, then we split the dataset. In the second strategy, we first split files, and then we created instances. Also, from the second strategy, we can compute two types of performances. Hence, we have performances based on instances (comparable to strategy 1) and based on files. Actually, one file is tagged by the corresponding class, which means that instances created from one file belong to this class. For example, if 10 instances on 13 are classified class A, then the file is predicted class A (it is a majority vote). It should be mentioned that only falls performed in position 1 have been used for strategies 1 and 2.

The third strategy is more sophisticated and allows evaluating the generalization of the proposed system operating in real conditions. To do so, we exploit the leave-one-subject-out method. The idea behind it is to select $N-1$ persons to represent the training dataset, and to use the Nth person as the testing dataset. In this study, $N = 10$.

Most information on the training of the CNN-LSTM models should be mentioned. First, the project has been developed with the open-source neural network library Keras and the integrated development environment PyCharm. Second, we used the optimizer that implements the Adadelta algorithm. It is an extension of the Adagrad algorithm, which is a stochastic gradient descent method. The Adadelta algorithm has the particularity to adapt the learning rate, while the model is training. Accordingly, the algorithm can adapt its learning with updated gradients. The initial learning rate is 0.001. Also, we defined a batch size of 256 instances. According to the computer (OS: Windows, RAM: 32 GB, Processor: i7-8850H, GPU: NVIDIA Quadra P3200) on which the training is done, it is a satisfying batch size since the tradeoff between the duration of the training (approximately 2 minutes and 30 seconds) and the performances (see after) of the model is excellent.

When we train neural network models, it is common to use a validation dataset. This dataset allows tracking if the model

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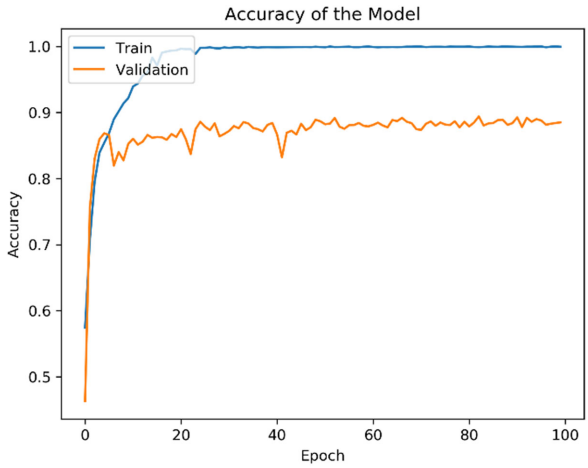


Fig. 6. Evolution of the accuracy vs. epoch during the training stage of the model related to the results presented in Table I.



Fig. 7. Evolution of the loss vs. epoch during the training stage of the model related to the results presented in Table I.

overfits the training data. In other words, the model continues to improve their performances on the training data while it gets worse on the validation and testing datasets. In this work, we used the testing dataset as a validation dataset to track the training of the model since we do not perform any fine tuning on the selected models. Also, all presented performances belong to classification models, which are not overfitted. Figs. 6 and 7 illustrate the evolution of the accuracy and the loss versus epoch during the training stage of the model related to performances given in Table I (for strategy 1). We observe that the accuracy stops to increase after several epochs. At the same time, the loss of the validation dataset starts to increase while the loss of the training dataset continues to decrease. This situation represents when the model starts to overfit (even if the loss value is low). Thanks to a callback function based on the evolution of the accuracy and loss values, we saved the weights and bias of the model before it falls into overfitting.

TABLE I
PERFORMANCES FOR FALL DETECTION BY TRAINING AND TESTING THE MODEL WITH THE STRATEGY 1

CNN-LSTM performances based on instances, Strategy 1.				
Global Accuracy	0.89			
Global Cohen's Kappa	0.77			
	P	R	F	Support
Non-Fall	0.91	0.88	0.90	394
Fall	0.86	0.89	0.88	314

TABLE II
PERFORMANCES FOR FALL DETECTION BY TRAINING AND TESTING THE MODEL WITH THE STRATEGY 2

CNN-LSTM performances based on instances, Strategy 2.				
Global Accuracy	0.85			
Global Cohen's Kappa	0.70			
	P	R	F	Support
Non-Fall	0.93	0.81	0.86	402
Fall	0.78	0.91	0.84	296
CNN-LSTM performances based on files, Strategy 2.				
Global Accuracy	0.87			
Global Cohen's Kappa	0.74			
	P	R	F	Support
Non-Fall	0.94	0.76	0.84	41
Fall	0.83	0.96	0.89	52

Finally, we proceed to a data augmentation for the training stage by creating 5 other instances from one instance. To do so, we permuted time-series TS_i, where $i = 1, 2, 3$. In other words, the instance denoted by $[TS_1, TS_2, TS_3]^T$ gives $[TS_1, TS_3, TS_2]^T$, $[TS_2, TS_3, TS_1]^T$, $[TS_2, TS_1, TS_3]^T$, $[TS_3, TS_2, TS_1]^T$, and $[TS_3, TS_1, TS_2]^T$. The main advantage of this technique is that the proposed system becomes independent of the localization of activities (events). Unfortunately, we did not perform classical data augmentation to increase the size of the dataset. The data augmentation task is very complex in this work and deserves more research. Indeed, the radar provides data that are arranged as a radar scattering matrix. The radar scattering matrix can be viewed as an image with only one channel. However, we have three UWB radars in the apartment, collecting the data of one action at three different locations. It is like taking a picture in three different angles of a subject performing a particular action. In this case, it is challenging to generate three different but coherent (for the same action) images (one for each angle) of the subject.

Tables I and II present the performances of our approach.

Performances in Table I represent fall detection efficiency according to strategy 1, which splits the dataset into two parts (70-30). It should be recalled that the present paper addresses a binary classification problem, where the Fall and Non-Fall events represent the positive and negative classes, respectively. The CNN-LSTM model, in this case, reaches 89% of global accuracy and 77% for Cohen's Kappa. In other words, for 394 instances belonging to the *Non-Fall* event, 347 instances are correctly classified (47 false positives). Concerning the *Fall*

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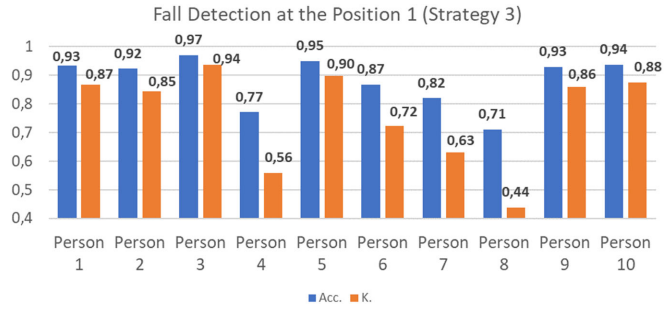


Fig. 8. Fall detection performances of the leave-one-subject at Position 1.

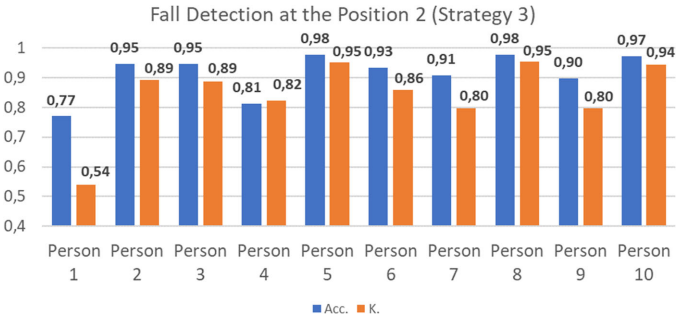


Fig. 9. Fall detection performances of the leave-one-subject at Position 2.

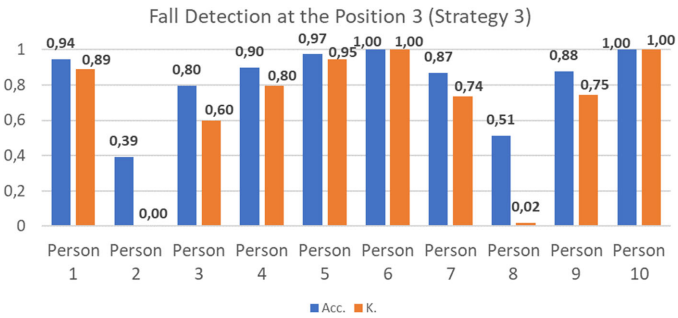


Fig. 10. Fall detection performances of the leave-one-subject at Position 3.

instances, 279 on 314 instances are detected as a Fall event (35 false negatives). As we can see, the performances are high, and only the false negatives of the *Fall* class is concerning.

In Table II, we can see that global performances are slightly lower than those in Table I. The difference can be explained by the Strategy 2, which splits the dataset into two parts (70-30), according to the number of files. Indeed, each *Fall* event can be unique, and, in this case, there is a possibility that one fall in the testing dataset does not share any similarities with all simulated falls in the training dataset. However, the performances of the CNN-LSTM model for instances are 85% and 70% for global accuracy and Cohen's Kappa, respectively. From a different point of view, 326 on 402 instances belonging to the *Non-Fall* class are correctly classified (76 false positives). Also, the model assigned the label *Fall* to 345 instances and 269 from those belong to that class. Concerning the number of false negatives for the class *Fall*, there are only 27 instances. In the second part of Table II, we compute performances based on the majority classification of instances per file (or per period of 20 seconds without overlap). From this point of view, the performance for fall detection is very high since 50 falls have been correctly identified over 52 falls.

For the last strategy (Strategy 3), we proceed by using the leave-one-subject-out. The method is simple. Indeed, in our study, *Fall* and *Non-Fall* events of 9 participants compose the training dataset, while instances of the 10th participant represent the testing dataset. Then, iteratively, each of the ten subjects is isolated to be the testing dataset. This method allows for observing the generalization performances of our solution. Figs. 8 to 10 illustrate results. It shall be noted that the average accuracy is very close to performances from Table I. Fig. 8 represents the simulation of falls at the position 1, we have 6 persons with scores superior to 90%, 2 participants with an accuracy between 80% and 90%, and 2 participants with a score lower than 80%. In Fig. 9, there are 8 persons with a global accuracy superior to 90%, 1 participant with an accuracy between 80% and 90%, and only 1 participant with a score lower than 80%. Finally, in Fig. 10, we observe many performance differences between participants. Indeed, the CNN-LSTM model performs efficiently with 5 persons since global accuracy is superior to 90%. Also, there are 3 participants with a score between 80% and 90%. More surprisingly, two of over ten participants have poor performances (lower than 50%).

Some reasons can explain these performance differences. First, simulated falls are different from one person to another. Thus, to increase the generalization of our models, we need many more participants. Secondly, walls, doors, and furniture can reduce the quality of the recorded data. It is the main explanation of the significant difference in the performances for position 3 (Fig. 10).

The performances presented in this paper are lower than those in the literature. However, it is difficult to compare adequately. Indeed, as you can see in Table III, most of the study exploits only one radar, and the conditions of the experiments depend greatly on the room configuration. For example, as presented in the related work, most of the work collected data in a small room (with or without furniture), where the participant is very close to the radar, increasing the detection rate. Furthermore, according to the study presented in [33], the number of participants, and the types of simulated ADLs play a significant role in the performances. Besides, in our study, the quality of data for the recording falls in the bathroom is affected by the presence of walls. Finally, it shall be noted that we detected falls in each position separately with data provided by the closest UWB radar. The obtained performances are comparable to those presented in the literature. Indeed, global accuracies (with the corresponding Kappa coefficient) are 98.51% (94.80%), 96.75% (90.88%), and 95.48% (87.39%) for positions 1, 2, and 3, respectively.

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TABLE III
PERFORMANCES OF THE LITERATURE VS. THIS STUDY

Works	Technologies, Number of exploited sensors	Algorithms	Number of participants	Number of Fall and Non-Fall events	Strategies	Accuracy
Sadreazami *et al.* [40]	UWB radar, 1	CNN	5	206 (121, 85)	TTS	92.72%
Erol *et al.* [14]	UWB radar, 2	SVM	6	∼400	TTS	97.88
Shrestha *et al.* [33]	FMCW radar, 1	SVM	6, 9	∼84, 180 (2 different datasets)	TTS	94%, 76.44%
Wang *et al.* [41]	UWB radar, 1	CAE, GRU	4	NA	TTS	96.80%
Li *et al.* [37]	(FMCW radar, 1) + (Inertial sensor, 1)	multi-layer bi-LSTM + Soft-hard fusion	16	NA	LOSO	96%
Our work	UWB radar, 3	CNN-LSTM	10	660 (600, 60)	TTS, LOSO	(98.5%, 96.7%, 95.5%), (88.1%, 91.5%, 82.6%)

TTS: Train/Test split; LOSO: Leave-One-Subject-Out.

VII. CONCLUSION

We proposed a new system to detect falls in an indoor environment by using UWB radars and a CNN-LSTM architecture. Results show high performances since we reached approximately a global accuracy of 90%. Also, compared to the literature exploiting UWB radars, it is the first time that such an in-depth study is performed. Indeed, experiments have been conducted in an apartment of 40 square meters, participants simulated falls at three locations, data are recorded from three UWB radars, and CNN-LSTM models have been tested with three strategies (Strategy 1, 2, and 3) including one that tests generalization of the proposed solution. Hence, according to this study, we can assert that UWB radars are an up-and-coming technology for fall detection and activity recognition.

However, some improvements can be made to increase the fall detection rate. Here, there are three perspectives: we should recruit much more participants; participants should simulate falls in more locations; we can explore other data pre-processing techniques such as those presented in the related work.

Moreover, one of the future works is to get performances of fall detection in real-time. Thus, we need to develop three identical wireless communication (e.g., Wi-Fi) devices that transfer the UWB data to a central system where the model is implemented. More specifically, each device embeds a UWB radar connected to a highly-integrated System-in-Module (e.g., Samsung's ARTIK 520). Each device will also integrate a power adapter that converts the power supply provided by the AC plug. Then, the highly-integrated System-in-Module gets the UWB radar data and transfers them to the central system. The central system could be a small-board computer (e.g., Raspberry Pi) that acquires, processes the data, and gives the classification with the implemented model. Further study is needed to make the right choice of components.

Finally, we are very enthusiastic (from the results obtained in our first exploration with raw data) to continue the research exploiting UWB radars.

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